

Voices in the Air: The Origins of Competition Policy

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Abstract

Where do competition-policy ideas come from, and how do they enter merger reasoning? We develop a measurement system that represents each theory of harm in a merger-policy document as a connected graph of directed mechanism claims—source node, relation, target node—rather than as a bag of topics or keywords. The policy corpus spans the Monopolies Commission of 1948 through the contemporary Competition and Markets Authority, covering substantive competition-harm reasoning from 1950 to 2025. For origins comparisons, the planned baseline source reservoir is the parallel *FrontierGraph* corpus of economics papers; an assembled extraction of 21 named antitrust-economics contributions from Coase 1937 to Baker 2019 provides an interpretable benchmark. The origins comparison now uses canonical triples: each policy theory of harm will be mapped into a small directed graph of ontology-resolved mechanism claims and compared with external mechanism graphs. The comparison asks how much policy reasoning is assembled from mechanism components already present in academic source corpora, which sources contain those components, and when the components first appear. The intended contribution is a measurable account of policy reasoning as an object with structure, persistence, and historical variation, together with a way to compare that reasoning with academic and legal source corpora. We treat these empirical objects as descriptive, not causal; the paper states what additional evidence would be needed to move toward causal claims about idea diffusion.

⟨placeholder⟩ Status note: the current draft makes the object, sample, and planned comparisons inspectable. The policy-side concept extraction is now complete. The main open empirical tasks are settling concept normalization, expanding the legal/practitioner source side, matching policy mechanisms to source reservoirs, and building the causal/outcome designs. Red notes mark substantive choices or work still needed; blue notes mark current empirical status.

1 Introduction

<placeholder> Settle the title and opening emphasis. Frame the paper around the substantive origins and consequences of competition-policy reasoning; the graph representation is the way we make those questions measurable, not the final object of interest.

Competition policy is often described as evidence-based and technocratic. But in practice, policy decisions travel through reusable ideas about how markets work, how mergers create harm, and what kinds of rivalry matter. Those ideas need not come directly from the current research frontier. They may instead come from older and more absorbed traditions: synthesis texts, teaching materials, legal reference works, and schools of thought that have become institutional common sense. If that is true, then the language of policy is not only a record of decisions. It is also a record of which ideas have become usable inside the state.

Merger control is a useful place to study this problem because authorities are forced to explain why a transaction may lessen competition. Modern decisions often do so through explicit theories of harm. They describe how a merger may weaken rivalry, foreclose access, raise barriers to entry, or alter innovation incentives. But two decisions can share broad vocabulary while relying on very different economic logic. A flat representation based on topics or keywords misses that distinction. The object that matters is the mechanism narrative itself.

This paper studies where competition-policy ideas come from and how they enter merger reasoning over long horizons. We begin from a simple empirical claim: policy reasoning should be measured at the level at which it is actually expressed. We therefore represent each theory of harm as a connected causal graph—a set of directed edges of the form (source concept, relation, target concept)—rather than as an unordered bag of terms. That object preserves sequence, direction, and internal structure. It also makes it possible to compare policy narratives to external source corpora.

The paper asks two empirical questions. First, what mechanism structure appears in long-run UK competition-policy reasoning? We apply the extraction procedure to a corpus spanning 1950 to 2025—Monopolies Commission and Monopolies and Mergers Commission inquiries, Competition Commission and Office of Fair Trading decisions, and modern CMA cases. This descriptive layer measures the number of theories of harm, their graph structure, and their mechanism-family labels before interpreting variation across institutional periods. Second, how much of that mechanism structure appears in academic and other source reservoirs? The comparison design maps policy concepts to a common economics ontology, converts each theory of harm into a canonical-triple subgraph, and compares those subgraphs with the full *FrontierGraph* academic corpus and with the named-contributions benchmark. **That comparison is not yet populated in this draft. Once concept normalization and source-reservoir matching are settled, the revised results will report how much of a policy theory of harm is assembled from academic mechanism components, which papers contain those components, and when the components first appear.**

The paper’s contribution is a measurable account of policy reasoning as an object with structure, persistence, and historical variation. The first contribution is a representation of competition-policy reasoning based on theory-of-harm graphs—a unit that has not previously been extracted systematically from the long-run UK corpus. Simpler objects used in earlier work, such as case-level outcomes, mechanism-family shares, and keyword-based measures, can be read as projections of this graph object: they keep some information while collapsing direction, sequence, or within-theory structure. The second contribution will be descriptive facts about the composition of those mechanism narratives, with the exact claims to be fixed after the measurement checks in Section 7. The third is a comparison design that can be applied to academic, legal, and practitioner source corpora, together with a clear boundary between descriptive overlap and causal claims about idea diffusion.

That boundary matters. A useful first step is to ask whether policy narratives look relatively more economics-like, more legal-like, or more mixed, and whether that balance changes across periods and outcomes—not to claim that any named thinker directly caused any given decision. The long-run corpus matters here because the phrase *theory of harm* itself is historically recent; earlier Monopolies Commission and MMC documents contain the same underlying mechanism logic in less standardized language. Part of the paper’s substantive agenda is to ask how competition-policy reasoning became formalized, not just which mechanisms appear most often.

The rest of the paper is organized as follows. Section 2 positions the paper against three prior literatures: computational measurement of policy reasoning, the intellectual history of antitrust, and the diffusion of ideas from research to policy. Section 3 explains why merger policy is a useful setting and how the long historical horizon changes the question. Section 4 defines the measured object and clarifies what this paper means by an idea trace. Section 5 describes the long-run policy corpus and the comparison corpora. Section 6 explains extraction, normalization, and validation. Section 7 will report descriptive facts about the extracted policy corpus once the remaining checks are complete. Section 8 turns to the origins question. Section 9 studies policy consequences and economic relevance. Section 10 states the limits of the current design. Section 11 concludes.

2 Related Literature

⟨placeholder⟩ Tighten around the papers that matter for the substantive question: merger-decision coding, antitrust intellectual history, and research-to-policy diffusion. Add only references that change how we position origins, timing, or policy consequences.

The paper sits at the intersection of three bodies of work, each of which it relates to but none of which it reproduces. The first is the growing literature that uses computational text analysis to measure the content of policy reasoning and scientific claims at scale. The second is the intellectual history and sociology of antitrust, a primarily narrative literature that reconstructs how ideas about competition have evolved in thought and practice. The third is the political-science literature on

the diffusion of policy ideas from research into regulatory practice. Situating the paper against all three is important because its contribution is partly methodological (what to measure), partly substantive (what the measurement reveals about merger policy), and partly about the limits of influence claims when the evidence is textual.

Computational measurement of policy reasoning and scientific claims. A first strand is the rapidly expanding literature that treats policy and scientific documents as structured data objects rather than as bags of words. Classic text-as-data methods in economics and political science have relied on topic models, term-frequency representations, and supervised classifiers (for an overview see Gentzkow et al., 2019). Recent work has moved toward extracting more structured objects directly: claims, relations, and mechanism chains. The most methodologically proximate precursor to this paper is Garg and Fetzer (2025), which extracts causal claims at scale from the economics research literature using large language models, producing a graph representation of claims that is comparable across documents and across fields. The present paper uses the same graph representation—directed edges between short node labels, a fixed relation vocabulary, and a connected narrative within each claim cluster—but applies it to a different corpus (UK merger policy) and asks a different question (where the policy claims come from, intellectually). The move from a research frontier to a regulatory corpus matters both substantively and methodologically: the documents are shorter, the register is different, and the claims are embedded in decisions rather than in abstracts.

A related, smaller literature studies the content of merger decisions quantitatively. Duso et al. (2024) document the evolution of theories of harm in EU merger control by hand-coding a sample of Commission decisions and characterising compositional change over time. Affeldt et al. (2021) extend this line to 25 years of European merger enforcement and report shifts in case composition and in the stated grounds for intervention. Duso et al. (2007) and Duso et al. (2011) evaluate the effectiveness of EC merger decisions against stock-market reactions and counterfactual outcomes. These papers are the closest quantitative neighbours to our project: they take the text of merger decisions seriously as data, and they operate at a comparable institutional scale. The distinction is one of granularity and primary object. Their unit of observation is usually the decision, the case, or the broad theory-of-harm category; ours is the theory-of-harm graph, with directed mechanism edges as its components. Their hand-coded categories can be read as coarser projections of the graph object: they preserve the broad family of harm but collapse the within-family chain, direction, and intermediate concepts. We read their results as complementary: they establish what changes in the mix of merger cases, theory-of-harm categories, and enforcement outcomes; we aim to establish what changes in the mix and internal structure of mechanism claims.

The intellectual history of antitrust and competition policy. A second strand is the intellectual history of competition-policy ideas, a mostly narrative literature that traces the succession

of schools of thought through which antitrust has been thought and taught. The canonical anchor for this paper is the Hovenkamp–Crane anthology *The Making of Competition Policy* (Crane and Hovenkamp, 2013), which organises the history of antitrust thought into twelve chronologically ordered chapters spanning classical political economy, progressivism, the new deal, ordoliberalism, the structural IO tradition, the Chicago School, and transaction-cost and post-Chicago synthesis. Our canonical source corpus and our tradition taxonomy both take this anthology as their starting authority, supplemented by named extensions for traditions (modern enforcement synthesis, post-Chicago game theory) that postdate or fall outside the anthology’s scope. Closely related narrative works include Kovacic and Shapiro (2000), which surveys “a century of economic and legal thinking” about antitrust, and the practitioner-facing textbook treatments of Motta (2004) and Whinston (2007), both of which are represented in our canonical corpus as primary sources and which have themselves shaped how competition authorities articulate harm. Baker (2019) offers a more polemical, modern-enforcement-oriented restatement of the field; it too is represented in our corpus.

This literature is essential background for us and, in a direct sense, data for us: the intellectual histories identify which texts count as canonical within each tradition, and several of them are themselves in our comparison corpus. But the narrative method in this literature does not produce the kind of document-level, claim-level evidence that would allow one to quantify the degree to which policy reasoning in any specific period reflects any specific tradition. The question “how much of contemporary UK merger reasoning is Chicago-School-like versus post-Chicago-like?” is not a question one can answer by reading Hovenkamp and Crane, because the anthology is a map of ideas in thought, not a measurement of ideas in policy use. The contribution of the present paper is to turn that map into a comparison instrument by extracting mechanism claims from policy documents and from source texts in the same representation, and asking how close they are in that representation.

Diffusion of policy ideas from research to practice. A third strand is the political-science and public-policy literature on the diffusion of policy ideas. Simmons et al. (2006) and Dobbin et al. (2007) develop a typology of diffusion mechanisms—social construction, coercion, competition, and learning—that has become the standard account of how policies travel across jurisdictions and across time. Shipan and Volden (2008) provide a parallel account for subnational diffusion in the United States. Earlier work by Weiss (1979) on the “many meanings of research utilization” and by Nutley et al. (2007) on how evidence is used in public-sector decision-making lays out the general problem of how research knowledge enters policy practice. Boswell and Smith (2017) and Hahn and Tetlock (2008) provide more recent treatments of the knowledge-policy relationship and the extent to which policy decisions actually incorporate research evidence.

Two things distinguish the present paper from this literature. First, the diffusion literature typically operates at the level of whole policies or policy regimes—did a jurisdiction adopt this

regulation, did a country enact this type of competition law—whereas our unit of observation is a single mechanism claim embedded in a single merger decision. The scale is deliberately finer. Second, the diffusion literature usually operationalises influence through adoption events and policy dummies, whereas the present paper operationalises resemblance through textual comparison in a structured representation. These two differences mean that the diffusion literature gives us the research question—ideas as objects that can travel from research into practice through identifiable channels—while the paper’s empirical strategy sits closer to the computational-text-analysis strand. In Section 10 we lay out a design space for causal claims about diffusion that would, in principle, let us test the mechanisms described by Simmons, Dobbin, Shipan, and Volden at the mechanism-claim level. This paper does not execute those designs; it provides the measured mechanism units on which they could be built.

The paper’s distinctive contribution relative to all three literatures. Relative to the computational text-analysis strand, the paper’s contribution is a new empirical object: a structured mechanism-chain representation of a long-run regulatory corpus, comparable to a structured representation of the intellectual traditions that might have shaped it. Relative to the merger-decision coding literature, the paper makes outcomes, broad families, and keyword measures recoverable as projections of a richer mechanism object. Relative to the intellectual history of antitrust, the paper’s contribution is turning a taxonomic map of schools of thought into a quantitative comparison instrument that can be applied to individual policy mechanisms. Relative to the diffusion-of-ideas literature, the paper’s contribution is shifting the unit of analysis from whole-policy adoption to mechanism-level co-occurrence, which would let later work run event-study designs at a finer scale than policy-regime adoption. The paper does not settle which of the Hovenkamp–Crane traditions “caused” the CMA’s vocabulary. It does show that UK competition reasoning can be represented as structured mechanism claims, mapped into a shared ontology, and compared to external reservoirs of academic and legal reasoning under explicit robustness checks. That is a narrower claim than the surrounding literatures sometimes make. It is also a more modest starting point for any subsequent attempt to make causal claims about the intellectual sources of modern competition-policy reasoning.

3 Institutional and Historical Context

(placeholder) Settle the baseline sample. The broad UK competition-harm corpus gives the long historical horizon; a merger-only corpus gives a cleaner institutional object but starts later. The paper should state which is primary and why.

This section explains why the UK corpus can support a long-run study of policy reasoning. It does three jobs: it motivates merger control as a setting where authorities must write down mechanism claims; it explains why the form of those claims changes across institutional periods;

and it defines the broad competition-harm sample used for historical analysis.

3.1 Why merger policy is a useful setting

The research question of the paper—where competition-policy ideas come from, and how they become usable inside the state—could in principle be studied in any policy domain where authorities are required to explain their substantive reasoning in writing. We focus on merger control for three reasons that are specific to the question, not to convenience.

First, merger cases impose an unusually consistent analytical object. Every merger review, regardless of sector or period, asks the same underlying question: if the transaction proceeds, will competition in a relevant market be harmed, and by what mechanism? Unlike abuse-of-dominance investigations (where the nature of the alleged conduct is open-ended) or market studies (where the remit depends on a sectoral referral), merger cases have a fixed shape. The authority is presented with a specific transaction, must delineate one or more relevant markets, must construct a counterfactual, and must assess whether the transaction makes the counterfactual outcome materially worse for competition. That uniformity of task is what makes it meaningful to compare a 1968 Monopolies Commission merger report with a 2022 CMA phase-two inquiry at all. They are different in almost every other respect, but they are answering the same type of question, and the rhetorical fingerprints of how each authority goes about answering it become the primary object of comparison.

Second, merger reasoning is forced to be explicit. Competition authorities that clear or block mergers are subject to judicial review and, in the UK case, to an active appellate body with the power to remit decisions. This procedural constraint makes reasoning part of the decision itself rather than discretionary commentary. A decision that declines to articulate the mechanism by which the transaction is expected to harm competition is, in contemporary practice, a decision that is vulnerable on appeal. From the researcher’s point of view this is a substantial advantage: merger documents are structured to reveal the authority’s substantive logic, not to hide it. When the authority believes the transaction threatens coordinated effects through increased information transparency in a differentiated-products market, the decision will say so, and will usually cite the evidence chain by which that belief is supported. That property makes merger documents one of the most attractive single sources of written policy reasoning available in any regulatory domain.

Third, the theory-of-harm framing is both institutionally prominent and analytically specific. Contemporary UK merger decisions often organise their substantive analysis around a numbered list of distinct *theories of harm*—horizontal unilateral effects, coordinated effects, vertical foreclosure, conglomerate effects, loss of potential competition, innovation effects, and so on—and assess each separately. The theory-of-harm framing is not merely a labelling convenience. It reflects the operational structure of the analysis: each theory is evaluated on its own mechanism story, its own evidentiary standard, and its own counterfactual, and the decision documents which theories are raised, investigated, upheld, or rejected. This structure gives a clean way to count and compare

substantive claims. It also, crucially, is not a universal constant of competition-policy practice. As we discuss below, the explicit use of the phrase *theory of harm* is historically recent. Part of the paper’s agenda is to ask what merger authorities did in the decades before they used that vocabulary, and whether the underlying mechanism structure can be recovered from the older documents by the same extraction procedure used for modern decisions.

The combination of these three features—analytical uniformity, required explicit reasoning, and a standardised contemporary framing that has nonetheless changed historically—is what makes merger policy a useful setting for the paper. An alternative setting that had any two of the three would be viable; a setting that had none of the three would not support the kind of measurement the paper is built around.

3.2 The changing form of harm reasoning

<placeholder> Separate real change in reasoning from changes in document form, statutory test, and extraction yield. Pre-SLC and modern documents may contain similar mechanisms, but they do not express them in the same institutional language.

The UK competition-policy regime has passed through four distinct institutional forms in the period covered by the paper’s long-run corpus. The Monopolies Commission, established under the Monopolies and Restrictive Practices (Inquiry and Control) Act 1948, was the first UK body with a sustained remit to investigate market-power concerns in specific industries; its approach was inquiry-led and focused on whether identified conduct operated against the *public interest*, a statutory test that explicitly blended competitive harm with social, political, and industrial considerations. The body was reconstituted as the Monopolies and Mergers Commission by the Fair Trading Act 1973, gaining a formal role in merger review alongside monopoly and anti-competitive practices investigations. From 1999, the Competition Commission took over substantive merger inquiry work, running in parallel with the Office of Fair Trading as the first-phase screening body. The Enterprise Act 2002 replaced the public interest test with a *substantial lessening of competition* (SLC) test, narrowing the statutory question to one of competitive effect alone except in a small number of specified public-interest cases. The CMA, established by the Enterprise and Regulatory Reform Act 2013, took over from April 2014 as the unified body handling both phases of merger review.

These institutional shifts are not incidental. They change what the authority is asking and how it is allowed to answer. Under the public interest test, competitive considerations sat alongside considerations of employment, regional development, export competitiveness, and industrial structure; a merger could in principle be cleared on grounds that had nothing to do with competition per se, and the reasoning in such a clearance would reflect that mixture. Under the SLC test, the statutory question becomes purely one of competitive effect. Reasoning that would have been written in terms of diffuse public-interest considerations in 1970 is, in 2020, written in terms of specific theories of harm with specific mechanism chains. The change is partly terminological—the

underlying substantive concern is similar—but it is also, and more importantly, structural. The set of things an authority is allowed to say has narrowed, and the set of things it is compelled to be explicit about has widened. This is the historical change that the descriptive-evolution section is meant to measure rather than assume.

The phrase *theory of harm* itself is a product of this institutional shift rather than a permanent feature of competition-policy practice. The phrase becomes standard in UK merger documents during the Competition Commission / OFT period, not earlier, and becomes nearly universal under the CMA. Earlier Monopolies Commission and MMC reports discuss competitive harm in more narrative and less numbered forms, often intermixing it with non-competition considerations, and rarely using the specific vocabulary of “theories of harm” or “mechanism stories.” The substantive content is not absent—reports from the 1960s discuss barriers to entry, market power, and cartel conduct in terms that are recognisable to a modern reader—but the representation is different in three specific respects: the reasoning is not numbered, the mechanism is not separated from the verdict, and the non-competition considerations are not partitioned off from the competition considerations. Recovering comparable mechanism structure from those earlier reports is a measurement problem as much as a substantive one.

The paper’s position on historical comparability is cautious but committed. We extract mechanism claims from pre-SLC documents using the same procedure we apply to the modern ones, on the working hypothesis that the underlying reasoning structure is present in the older documents even when the labels differ. We are cautious about the resulting statistics: systematic differences in extraction yield, edge counts per inquiry, or the mix of mechanism families between periods may reflect changes in how authorities wrote about harm as much as changes in the underlying substantive concern. We address this by reporting statistics separately for each institutional period where systematic differences exist, and by running the style and surface-feature audits described in Section 6.4. The long historical horizon is therefore a question to be investigated rather than a fixed baseline.

Two things are comparable across the full horizon. First, the substantive question of whether a transaction may harm competition is the same question in every period, even when the statutory test is formally broader. Second, the individual mechanism claims that make up a theory of harm—“entry barriers are higher post-merger because of X ,” “coordinated conduct becomes easier because of Y ,” “the merged firm’s control of Z enables customer foreclosure”—are the kind of sentences that appear, in recognisable form, in reports from every period in the corpus. Two things are not comparable. First, the overall case framing differs: a 1970 inquiry that blends competitive and non-competitive considerations does not have the modern numbered theory-of-harm structure even after extraction, and aggregating it into the modern taxonomy produces a category that the original document does not make. Second, the set of acceptable reasoning grounds has changed, so the absence of a category in the earlier documents does not imply that the authorities of that period would have rejected the underlying argument; it may simply mean the argument was organ-

ised differently or expressed in different language. We return to both of these distinctions when interpreting the long-run descriptive results in Section 7.

3.3 Object of analysis

<placeholder> Clarify the pooled object. If early monopoly inquiries remain in the main sample, describe them as competition-harm reasoning that later becomes merger-theory reasoning, with merger-only estimates as the clean institutional comparison.

The Monopolies Commission of 1948–1973 had no merger jurisdiction; its remit was monopolies and restrictive practices. The MMC of 1973–1999 added merger review on top of that earlier remit. By 2014 the CMA inherits a merger-focused practice, with monopoly-flavoured work routed through market studies and abuse-of-dominance investigations. A long-run corpus that limits itself to merger inquiries therefore begins in 1973 and excludes the entire pre-MMC period—roughly fifty inquiries of the 1950s and 1960s in which the same mechanism vocabulary (entry barriers, vertical foreclosure, market power, conditions of entry) was developed and applied to monopoly conduct rather than to prospective transactions. The paper takes the broader frame and treats *competition-harm reasoning* as the primary unit, including monopoly inquiries from 1948 onwards alongside post-1973 merger inquiries. The justification is straightforward: the procedural object differs across the two inquiry types (existing monopoly versus prospective merger), but the underlying mechanism claims do not. A 1957 inquiry asks whether concentration in a sector raises prices through restricted entry; a 2022 inquiry asks whether a transaction would do the same. At the level of directed edges between conceptual nodes, those are the same kind of claim. A separate *merger-only* subset can therefore be reported as a robustness cut where pooled historical comparisons depend on the broader competition-harm corpus. The two cuts should converge in the modern period and diverge mechanically in the pre-1973 sample, where the merger-only cut is empty by construction.

3.4 Cross-jurisdictional comparators

<placeholder> Use this only to motivate later causal designs and source expansion. EU and US material become central once we add guideline shocks, Brexit-related jurisdictional variation, or comparable foreign policy corpora.

The UK regime sits inside a wider merger-control network, especially through DG COMP and US antitrust enforcement. Those links matter for later identification designs: EU jurisdictional thresholds around Brexit, US and EU merger-guideline changes, and the parallel adoption of more economics-based merger analysis all create possible comparison settings. They are not needed for the current descriptive results. The present paper therefore treats EU and US corpora as comparison corpora for later work and keeps the main analysis anchored in the UK corpus.

4 Conceptual Framework: Ideas, Policy Reasoning, and Measured Objects

⟨placeholder⟩ Keep the hierarchy clear. The substantive unit is the theory-of-harm graph; canonical triples are its components; outcomes, mechanism families, and keyword measures are lower-dimensional projections used for specific empirical questions.

This section defines the object being measured. The paper uses “idea” in a narrow sense: a directional mechanism claim that can be found in policy text, represented as part of a theory-of-harm graph, and compared with mechanism claims in other corpora. That definition is deliberately narrower than a school of thought or a policy ideology, because the empirical analysis needs units that can be counted, dated, and matched.

4.1 What is an idea in this paper?

⟨placeholder⟩ Make sure the definition of “idea” serves the origins question. A mechanism claim is useful because it can be dated, compared across corpora, and linked to policy outcomes; avoid making the section sound like measurement is the contribution by itself.

The word *idea* is used loosely in discussions of policy intellectual history. It can refer to a named school of thought, a keyword or phrase in a document, a named author’s contribution, a mechanism, or a rhetorical frame. For the purposes of this paper we need a narrower and more operational definition, because we intend to measure ideas, count them, compare them across documents and across corpora, and track how they change over long time horizons.

We define an *idea*, in the primary sense used in the paper, as a *mechanism claim*: an assertion that some feature of a market or transaction bears a specific directional relationship to a specific outcome of interest. “Entry barriers are higher post-merger because the combined firm controls an essential input” is an idea in this sense; it is a directional claim about the relationship between two identifiable conceptual nodes. “The merger was blocked” is not—it is a decision, not a mechanism claim. “The transaction is in a concentrated market” is an idea only in the trivial sense of asserting a factual attribute; on its own it does not claim that anything causes or constrains anything else.

Mechanism claims are the component unit of the paper. The primary policy-reasoning object is the theory-of-harm graph, and four analytical units are built around it:

1. The *edge* is the measured form of a single mechanism claim. It is a directed triple (s, r, t) in which s and t are short text nodes and r is a relation category drawn from a small fixed vocabulary (*causes, increases, decreases, enables, constrains, associated with*). The normalization strategy discussed in Section 6.2 gives each edge a canonical form (c_s, r, c_t) , where c_s and c_t are canonical node identifiers. Edges are the components used inside both descriptive and origins analyses: when we ask whether a 2020 CMA claim resembles something in a 1985 academic text, we ask whether components of the policy theory-of-harm graph match mechanism

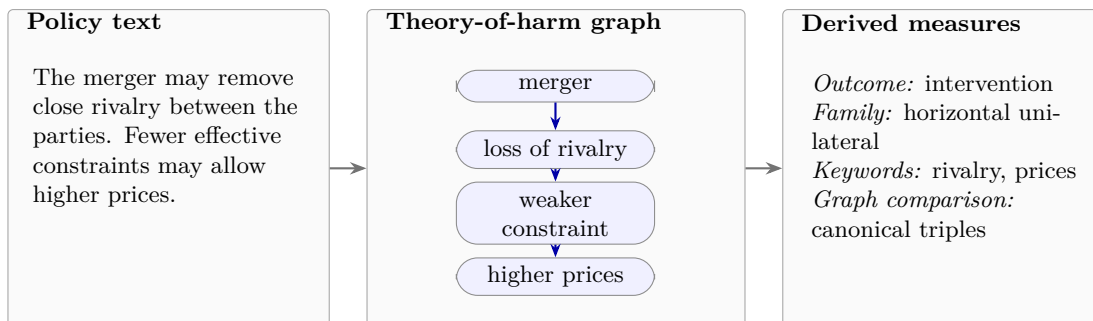
components in the shared ontology.

2. The *theory of harm* (ToH) is a small connected graph of edges that together tell one coherent mechanism story about how a merger may harm competition. A single theory of harm is a short narrative with internal structure: each node plays a role, the edges connect those roles into a causal or enabling chain, and the chain terminates in a stated competitive outcome. The extraction procedure (Section 6.1) requires each theory of harm’s edges to form a single connected component; disconnected content is split into separate ToHs. Theories of harm are the natural unit for case-level reporting, because a single merger case typically articulates a handful of distinct mechanism stories.
3. The *motif* is a recurring subgraph of edges that appears across multiple theories of harm, possibly across different cases and different corpora. A dyadic motif is a single canonical edge (canonical source $\rightarrow$ relation $\rightarrow$ canonical target); a triadic motif is an ordered pair of connected canonical edges; longer motifs are defined analogously. Motifs are defined at the canonical-node level, so two surface-different edges that share a normalized form count as the same motif. Motifs are the natural unit for tracking intellectual innovation: a new motif appearing in policy documents after a specific academic publication is the kind of diffusion signal the paper eventually wants to measure.
4. The *family* is a coarser grouping of edges or motifs under a shared mechanism-family label. Each theory of harm carries a top-level mechanism-family label drawn from a fixed twelve-category list—horizontal unilateral effects, coordinated effects, vertical foreclosure, conglomerate and portfolio effects, loss of potential competition, innovation effects, platform and network effects, data-advantage, bidding and procurement effects, entry and expansion, buyer power, and other—and a finer-grained leaf mechanism label. The top-level label is shared with the CMA Merger Assessment Guidelines vocabulary, which makes family-level counts interpretable to practitioners. Families are the natural unit for long-run composition analysis, where the research question is whether the mix of mechanism categories changes over time.

These four units are nested and complementary, not alternatives. Different analytical questions call for different primary units. Descriptive evolution of the composition of reasoning over time (Section 7) is primarily a family-level question. Cross-corpus idea tracing (Section 8.1) is primarily a theory-of-harm graph question, with canonical triples as the comparison components. Survival and entry of specific named mechanisms is a motif-level question. Case-level reporting to readers who want to know what a given merger inquiry actually argued is a theory-of-harm-level question. The extraction produces all four simultaneously, and each analysis uses the relevant unit.

This nesting is also the paper’s measurement claim. A case-level outcome measure observes the final decision but discards the mechanism. A family-level measure observes the broad channel of harm but collapses the internal chain. A keyword or embedding measure observes vocabulary but

Figure 1: A theory of harm as a directed mechanism graph. This is an orientation figure rather than a result figure. It shows how a policy-text passage is represented as a connected graph of mechanism claims, and how case outcomes, mechanism-family labels, keyword measures, and canonical-triple comparisons are lower-dimensional summaries of the same object.



not direction. The theory-of-harm graph keeps these lower-dimensional objects recoverable while preserving the direction and connection between mechanism components. The purpose of the graph representation is therefore not to add complexity for its own sake, but to measure the object from which the simpler quantities can be derived.

4.2 Why graph objects rather than flat text features

<placeholder> Shorten if needed. Keep only the parts that show why graph structure matters for substantive claims about origins, persistence, and policy consequences.

An alternative measurement strategy would represent each document by flat text features—term frequencies, topic-model loadings, or document embeddings—and compare documents in that representation. This strategy is standard in the text-as-data literature and is attractive for its simplicity and scalability. We do not adopt it as the primary representation for this paper, for three reasons that are specific to the subject matter.

First, policy reasoning in merger control is directional by construction. A sentence of the form “high entry barriers make coordinated effects more likely” asserts that one feature (entry barriers) increases the likelihood of another (coordinated effects). A flat representation treats that sentence as a joint occurrence of the two phrases, with no representation of which one is the source and which one is the target. But reversing the direction is not a rephrasing, it is a different theory. The claim “coordinated effects indicate high entry barriers” is also grammatical and also a merger-analysis statement, and any comparison method that cannot distinguish it from the original will treat intellectually distinct arguments as the same. Direction-preserving representations are therefore a necessary condition for the kind of cross-document comparison this paper wants to perform.

Second, the unit that matters for comparability is a chain of claims, not an individual phrase. Two merger cases can both use the phrases “market definition,” “entry barriers,” “concentration,” and “foreclosure” with identical frequencies and still be making very different arguments. The dif-

ference lies in how those pieces are connected—whether the case builds a horizontal unilateral story running from product substitutability through diversion to upward pricing pressure, or a vertical foreclosure story running from essential-input control through downstream competitor disadvantage to customer-side harm. Flat features cannot distinguish these because the distinction is in the connectivity of the mechanism nodes, not in the set of nodes themselves. A graph representation keeps the connectivity explicit.

Third, the economics of merger analysis has always been structured as chains of causal claims, even when the language used in the documents does not flag them as such. Merger assessment, whether conducted in 1970 or 2020, asks the same general question in a specific way: if the transaction proceeds, what is the mechanism by which competition will be harmed, and is there enough in the evidence to believe it? The answer has the form “because of X , Y will occur, which in turn will lead to Z .” The graph representation is a faithful formalization of how the substantive analysis is already organized. A flat representation is a lossier approximation, and for this subject matter the lost material is exactly the material that carries the economic argument.

We are not arguing that graph objects should replace flat text features for all purposes. Flat features remain useful as a validation check on the graph representation—for instance, for checking that families with increasing frequency over time also show rising term-frequency signals of their expected surface vocabulary. The claim is narrower: for the primary research questions of this paper, the measurement unit has to be the mechanism chain, and the mechanism chain is a graph.

4.3 Idea tracing

⟨placeholder⟩ Develop the distinction between overlap, precedence, and influence. The next empirical pass should show when components appear in source reservoirs and policy documents; causal influence requires additional designs rather than softer wording.

The measurement strategy produces, for each document in the policy corpus and in each source corpus, a set of mechanism-chain graphs expressed in a common representation. What does it mean to *trace* an idea from a source corpus to a policy corpus in this representation?

We use *idea tracing* as an umbrella term for a family of comparison questions that differ in ambition. The weakest form is *descriptive resemblance*: given a body of policy reasoning and a body of source-corpus reasoning, how much mechanism content overlaps across the two, and which parts of the source corpus account for that overlap? This is the question Section 8.1 will answer once concept normalization and source-reservoir matching are fixed. A stronger form is *diffusion dating*: for a specific mechanism or motif that appears in policy reasoning, when and where does it first appear in the source corpus, and does its policy frequency increase after specific source-corpus events such as publication of an influential paper or adoption of a new merger-analysis tool? This is a descriptive question with a time dimension, not a causal question, but it moves closer to the kind of diffusion narrative that intellectual historians of the field would want to tell. A still stronger form is *causal influence*: did a specific academic contribution *cause* a shift in policy reasoning, in

a sense that would let us write down a counterfactual in which that contribution did not occur? Causal-influence claims require event-study designs, shocks, or natural experiments, as discussed in Section 10.

This paper’s design is pitched at descriptive resemblance and, to a limited degree, at diffusion dating. It does not attempt causal claims about intellectual influence. We keep this distinction visible throughout the paper because the risk of over-reading similarity as influence is substantial when the comparison object is intellectual content: any finding that says “CMA reasoning resembles the Chicago School” will be tempting to read as “CMA reasoning was caused by the Chicago School,” and that inference is not supported by the method. The method is intended to support the weaker claim that components of UK competition reasoning can be matched to external reservoirs of mechanism claims, and that the timing and source distribution of those matches can be described.

Two clarifications about what similarity does and does not mean in this paper are worth making explicit. First, *directionality is preserved* in the comparison: a policy edge and a source edge are close only if both claim the same directional relationship between corresponding canonical nodes. A policy claim that “concentration increases coordination risk” is not treated as similar to a source claim that “coordination risk is a consequence of low concentration,” even though both sentences mention the same pair of concepts. Second, *the comparison is asymmetric*: a policy theory of harm may be larger than any one academic-paper graph, so we report what fraction of the policy graph is contained in the academic reservoir separately from what fraction of a paper-local graph is contained in the policy corpus. Time stamps then let us interpret matched triples as earlier or later appearances of the same mechanism component.

The remainder of the paper sets out how these definitions will be operationalised. Section 7 reports descriptive facts about the extracted policy corpus in the mechanism-chain representation. Section 8.1 sets out the origins comparison to be populated once concept normalization and canonical-triple matching are complete. Section 10 lays out what the paper would need to add to move from descriptive resemblance toward causal influence claims, and what that would require in terms of data and design.

5 Data

<placeholder> State the source hierarchy and expansion plan. FrontierGraph is the main academic reservoir for now, but the source side should be expanded with legal, practitioner, competition-policy, and antitrust-specific academic material that may be thin or missing in the current reservoir. Named contributions remain an interpretable benchmark.

This section separates the texts whose reasoning we explain from the texts used to interpret possible origins. The policy corpus is the object of study. The academic, legal, practitioner, and policy-source corpora are comparison reservoirs. Keeping that hierarchy clear is important: a source corpus can help locate where a mechanism appears, but it is not itself the policy outcome.

5.1 Overview

The empirical design separates the policy object from the source reservoirs used to interpret it. The baseline policy object is the long-run UK competition-harm corpus; a merger-only subset can be used as a robustness cut where historical pooling matters. The baseline academic source reservoir for origins comparisons is *FrontierGraph*; the assembled named-contributions corpus supplies a smaller, datable, interpretable benchmark. Table 1 reports the corpora used in this version. Additional corpora for cross-jurisdictional and legal-reference comparisons are described in Appendix A.

Table 1: Data overview: active corpora

Corpus	Source	Period	Docs/cases	Substantive	Extraction status
Historical inquiries (MC + MMC)	CC archive, digitised holdings	1950–1999	350	330	Concept extraction complete; comparability checks pending
CC “new” inquiries	CC archive (gov.uk archive)	2003–2014	137	105	Concept extraction complete; comparability checks pending
OFT phase-1 decisions	OFT archive (gov.uk archive)	2003–2014	417	146	Concept extraction complete; substantive-content filter pending
Modern CMA cases	CMA case register	2014–2025	774	758	Concept extraction complete; outcome coding pending
<i>UK subtotal</i>		<i>1950–2025</i>	<i>1,678</i>	<i>1,339</i>	<i>Concept extraction complete; final sample cut pending</i>
Named contributions	21 antitrust-economics works	1937–2019	21	21	Interpretable benchmark
FrontierGraph academic reservoir	Parallel economics-paper graph	various	~228,000 papers	—	Baseline academic reservoir

Notes: “Substantive” = documents that contain analytical reasoning about competitive harm, as opposed to procedural notices or jurisdictional non-qualification decisions. The OFT row shows that 271 of 417 OFT decisions are short jurisdictional notices without substantive SLC analysis. Paper-facing ToH and edge counts now come from the concept-level policy extraction, with final reporting pending sample-cut and comparability choices. The 1980s gap is documented in §5.2.

The UK corpus supplies the descriptive results. *FrontierGraph* is the baseline comparison corpus for the origins analysis. The named-contributions corpus provides datable, interpretable antitrust contributions that can be used as a benchmark and as case-study evidence, but it is not the main academic reservoir.

5.2 Long-run UK competition policy corpus

The primary policy corpus consists of substantive UK competition-authority documents spanning 1950–2025. It is broader than modern merger review because the pre-1973 Monopolies Commission had no merger jurisdiction; early monopoly inquiries are retained when they contain the same kind of competition-harm reasoning later expressed through merger theories of harm. Cartel investigations,

market studies, and abuse-of-dominance cases are excluded from the main analysis.

The current case index contains 1,678 documents, of which 1,339 contain substantive mechanism reasoning under the present substantive-content filter. Paper-facing ToH and edge counts now come from the concept-level extraction and will be reported after the sample cut and historical-comparability checks are fixed. The main gap is the 1980s: the digitised archive jumps from 1976–1979 cases to 1990 cases, with the intervening years held only as catalogue entries at the National Archives. The paper treats that decade as a documented gap rather than as a period with no competition-harm reasoning.

5.3 Source corpora for origins comparisons

The named-contributions corpus captures datable antitrust-economics ideas that are plausibly usable inside policy institutions. It contains 21 works from Coase (1937) through Baker (2019), selected from Hovenkamp and Crane’s intellectual history and supplemented with later post-Chicago, merger, and neo-Brandeisian contributions. The corpus is not a catalogue of antitrust thought. It is a structured benchmark whose provenance is reported in Appendix B. [For this draft, the named-contributions corpus is best treated as an interpretable source-side benchmark, not as the primary academic reservoir.](#)

The broader academic comparison comes from *FrontierGraph*, a parallel corpus of approximately 228,000 economics papers with about 1.24 million canonical-resolved edges and a 153,800-node ontology. This reservoir is broader than a merger-specific paper pull and lets policy mechanism components match adjacent literatures when the same canonical triples appear outside antitrust or merger venues. The project-specific OpenAlex merger-topic pull is therefore parked as a possible top-up rather than treated as the primary source corpus.

6 Measurement

<placeholder> Use the concept-level extraction for paper-facing ToH counts, edge counts, mechanism composition, and outcome summaries. Earlier extraction outputs are for quality control only.

This section describes how written reasoning becomes an empirical object. It covers three steps: extracting theory-of-harm graphs from documents, mapping surface phrases to common concepts, and comparing policy graphs with source-corpus graphs. The section also states the validation checks needed before historical or origins comparisons are interpreted substantively.

6.1 Extraction of theories of harm

The paper requires a representation that can be read by economists and compared across corpora. The extraction therefore converts each document in the policy and source corpora into a structured

set of theories of harm (ToHs), each of which carries a narrative summary, a sequence of narrative steps, and a set of directed edges over short text nodes. Inputs are decision texts, inquiry reports, book chapters, articles, and contribution paraphrases. The same extraction procedure is used for both.

The paper-facing policy extraction uses a concept-level representation. Earlier full-corpus extractions are useful for orientation and quality control, but their less stable node labels are not the right basis for the final origins, mechanism-composition, or outcome analyses. [The policy-side concept extraction is complete. Paper-facing canonical-triple results and outcome summaries are pending until concept normalization, source-reservoir matching, and outcome coding are settled.](#)

Each document is split into overlapping text chunks and submitted to a language model under a structured schema. For each theory of harm, the model returns a short narrative, ordered mechanism steps, directed edges between node phrases, evidence snippets, mechanism-family labels, and assessment status. For the modern CMA corpus, extraction was carried out using a multi-agent LLM workflow. Claude Opus 4.7, run with high reasoning effort, acted as the coordinating agent. Parallel Claude Sonnet 4.6 agents, also run with high reasoning effort, processed document batches under the same structured extraction schema. Each subagent returned JSON outputs conforming to the theory-of-harm graph schema. A deterministic post-processing layer removes empty nodes, normalizes relation labels, deduplicates repeated edges, and checks whether the edges inside a ToH form a connected mechanism story. When a document yields duplicate ToHs across chunks, a conservative within-case de-duplication step collapses near-identical ToHs. The completed policy-side run produced 2,806 theory-of-harm graphs with no bad JSON files, no schema failures, and no empty graphs after validation.

The same extraction approach is applied to historical reports that predate the phrase *theory of harm*. No historical prompt is used. The premise is that older reports often contain mechanism reasoning even when they do not organize it in modern numbered ToHs. Historical comparability is therefore an empirical validity question, not an assumption; Section 6.4 summarises the checks and Appendix C records the protocol.

6.2 Normalization and ontology

The extraction returns, for each document, a set of directed edges of the form (s, r, t) , where s and t are short free-text node labels and r is a relation label drawn from a fixed controlled vocabulary. The same underlying concept can surface under many different surface strings. A 1948 Bain essay on “price and production policies” may write *barriers to new entry* or *conditions of entry*; a 2022 CMA phase-one decision may write *barriers to expansion*, *countervailing entry*, or *potential competition from new entrants*. Cross-corpus comparison and motif survival therefore require node normalization.

The planned origins analysis uses the *FrontierGraph* ontology as the shared concept space. Extracted node strings are assigned to common concepts using a transparent sequence: exact

matches, known aliases, and embedding-based nearest matches. The main analysis will use only high-confidence concept assignments. Lower-confidence assignments will be retained for robustness and residual inspection. Unmatched policy nodes form a residual tail that can be inspected for recurring legal or regulatory concepts not already covered by the economics ontology.

Given canonical node identity, each extracted edge becomes a canonical triple (c_s, r, c_t) . Connected pairs or chains of such triples define motifs. The descriptive sections can use these motifs to study mechanism composition and survival over time, while the origins section will use them to compare policy theories of harm with source-corpus mechanism claims.

6.3 Comparing policy to source corpora

The origins analysis treats the policy theory-of-harm graph as the object to be explained and asks how much of that object appears in external source reservoirs. The comparison operates on canonical-triple subgraphs. Each extracted edge is first converted from a surface-text claim (s, r, t) into a canonical triple (c_s, r, c_t) , where c_s and c_t are ontology identifiers and r is the relation category. Each theory of harm is then represented as a small directed multigraph of these triples.

The main comparison is asymmetric containment. For a policy theory of harm P and an academic graph A , containment from policy to academic is

$$\text{Contain}(P \rightarrow A) = \frac{|P \cap A|}{|P|},$$

with the intersection computed over canonical triples. The reverse direction, $\text{Contain}(A \rightarrow P)$, will be reported separately because academic-paper graphs are typically smaller and serve a different interpretive purpose. Policy-to-academic containment will measure how much of a policy mechanism is found somewhere in the academic reservoir; academic-to-policy containment is closer to influence tracing.

Matches will be reported under three strictness levels. Level 1 matches canonical node pairs and ignores relation type. Level 2 adds directionality. Level 3 additionally requires relation-category alignment through the crosswalk table. Where exact canonical IDs do not match, the scoring rule allows continuous tolerance through the ontology hierarchy: exact matches receive weight 1.0, shared-parent matches 0.7, shared-root matches 0.3, and unrelated pairs 0.

The baseline origins specification uses the full *FrontierGraph* academic corpus as the source reservoir and reports policy-to-academic containment under the chosen strictness level. The named-contributions corpus remains useful because its works are datable and interpretable, but the broad academic reservoir is the main source for assembly, per-triple provenance, and origination-time results.

<placeholder> Settle weighting across theories of harm. Options include equal ToH weights, within-ToH averaged component weights, path-length weights, or narrative-step weights. Report both component-level and ToH-level results until we know how much the weighting choice matters.

⟨placeholder⟩ Empirical next step — ToH template entity resolution and aggregated graphs. Using the current concept-level extraction, cluster similar theory-of-harm graphs into recurring ToH templates. The existing mechanism-family label is only a coarse classification; it does not identify when differently worded ToHs instantiate the same mechanism. For each resolved template, aggregate its canonical triples into a weighted graph, with edge weights based on the number or share of ToHs/cases using that mechanism component. This would let the descriptive section report recurring mechanism graphs within broad families, and would let the origins section ask whether whole ToH templates, not only individual triples, appear in the academic reservoir.

6.4 Validation and historical comparability

The current results depend on two forms of measurement validity. First, the extraction must recover mechanism claims that are present in the underlying documents without adding claims that the text does not support. Second, historical and modern documents must be comparable enough that changes over time are not just changes in document style.

The current audit evidence is strongest for the modern extraction layer. A 200-edge audit sample from the CMA pilot found a fabrication rate below two percent. Against human-coded mechanism claims, precision is high and recall is moderate, with missed claims concentrated in long, multi-mechanism documents. The historical-comparability audit is still pending; until it is complete, pooled historical-modern statistics should be interpreted as descriptive and checked separately by institutional period.

The origins results add further dependencies. They require valid concept-level node normalization, meaningful cross-corpus comparison between policy and academic texts, and checks that paraphrase-based named-contribution extractions behave like primary-text extractions. Those issues affect the paper-facing descriptive counts, outcome summaries, and origins results. Appendix C records the current validation protocol.

7 Descriptive Evolution of Competition-Policy Reasoning

⟨placeholder⟩ Use this section to establish the main descriptive facts about competition-policy reasoning before the origins results. Start with the number of extracted theories of harm, their mechanism-family composition, and their structure by institutional corpus. Historical claims should be checked by inquiry type, case type, report length, and extraction consistency.

Current empirical status: the concept-level policy extraction covers 1,301 UK competition-policy cases with at least one extracted theory of harm, 2,806 theories of harm, 12,081 directed edges, and 19,339 extracted concept nodes. This replaces an earlier extraction whose node labels were less stable for cross-document comparison. The current ontology mapping combines high-confidence *FrontierGraph* matches with a first competition-policy extension. In the subset of relationships whose endpoints are resolved for origins analysis, 2,215 core-effect edges have both endpoints re-

solved. The descriptive objects below are therefore useful for inspection and sample-definition choices, but remain provisional until the ontology mapping and historical-comparability checks are settled.

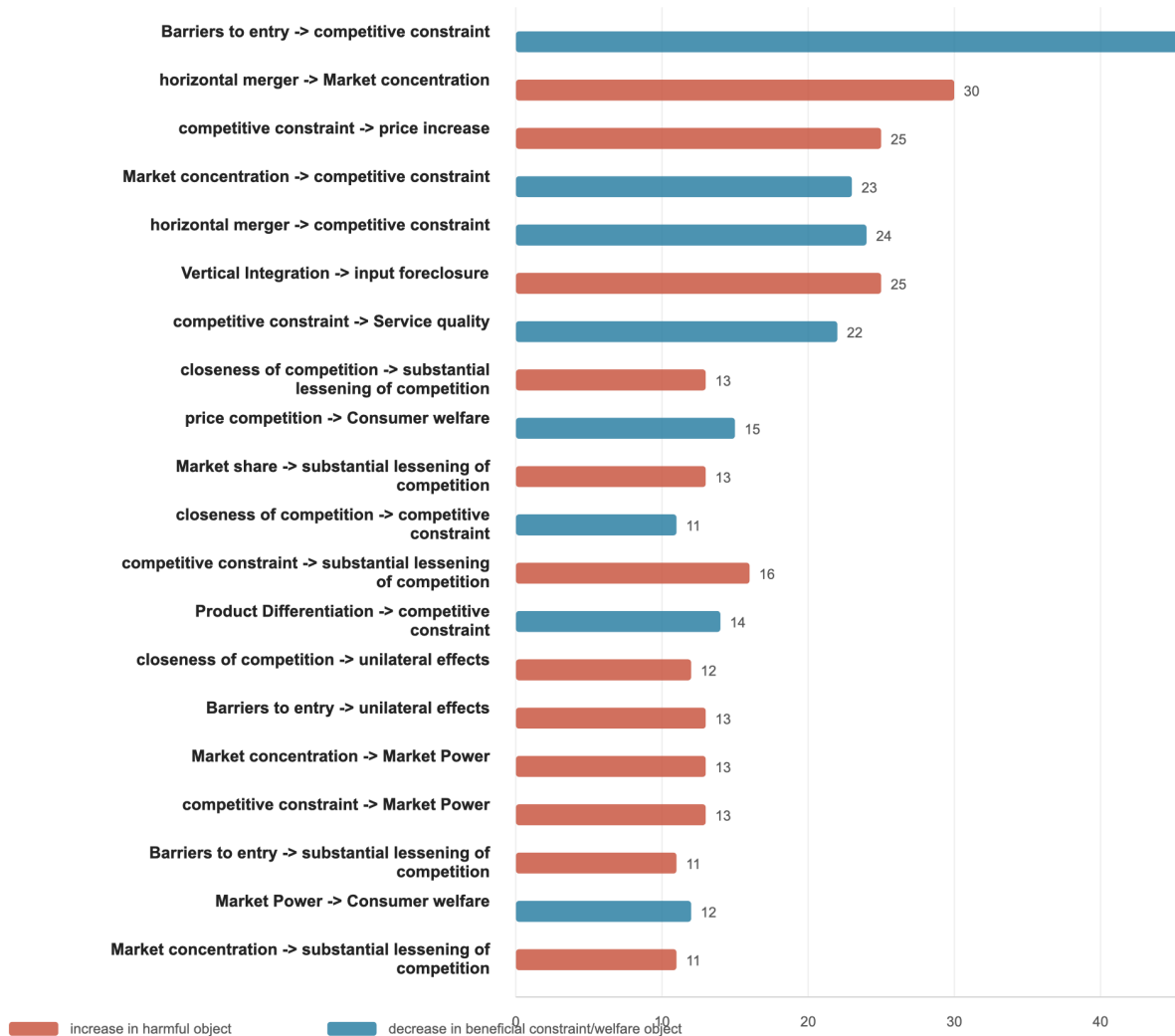
The descriptive section has a limited role in the paper. It shows what has been extracted from the policy corpus before asking where those mechanisms came from or whether they affected outcomes. The aim is not to rank the most important harms in UK policy. It is to establish the empirical object: how many theory-of-harm graphs we observe, how much of the graph can currently be resolved to common concepts, and what the first sign-aware mechanism relationships look like.

The first descriptive object is a sign-aware directed relationship between two resolved policy concepts. For example, an edge can say that a mechanism increases a harmful object, such as market power, or decreases a beneficial object, such as competitive constraint. This representation is narrower than the full extracted graph, because it only keeps edges whose endpoints can be mapped to resolved concepts in the current origins-analysis ontology. It is also more useful for the paper than raw edge counts, because it separates affirmative harm mechanisms from relationships that are mitigating, diagnostic, or not yet classifiable.

Figure 2: Common sign-aware mechanism relationships recovered by the extraction. Bars report the number of cases in which an extracted directed concept-edge appears. Orange bars denote edges that increase a harmful object; blue bars denote edges that decrease a beneficial object. The figure reports the top 20 aggregated concept-edges from 2,806 extracted theories of harm. Endpoints are resolved using the current ontology mapping.

Common sign-aware mechanism relationships

Resolved concept-edges; bars show number of cases. Color records the sign of the harmed object.



Cases per aggregated concept-edge
Sample: top 20 affirmative-harm aggregated concept-edges from 2,806 extracted ToHs; endpoints resolved using the current ontology mapping. Inspection figure.

Figure 2 is an inspection result: it shows the most common sign-aware mechanism relationships recovered by the extraction. It should not yet be read as a ranking of the most important harms in UK competition policy. Its role here is to make the graph object tangible. The recovered relationships are not simply topic labels; they are directed mechanism components, such as barriers to entry reducing competitive constraint, horizontal mergers increasing concentration, and vertical

integration increasing input foreclosure. The blue bars should not be read as benign relationships. They are affirmative harm paths in which the harmed object is a beneficial constraint or welfare object that falls.

Table 2 reports the provisional two-layer sample cut used in this section. The first row treats the long-run UK competition-harm corpus as the main historical object. The following rows separate the modern CMA subset from the non-CMA and historical remainder, because modern CMA decisions use more consistent theory-of-harm language and support cleaner status and outcome coding. The table is descriptive: it reports the size of the extracted object, the current rate at which directed edges can be mapped to resolved concepts, and the classification of resolved core-effect edges into affirmative harm, mitigating/no-concern, or still ambiguous relationships.

Table 2: Provisional descriptive status of the concept-level policy extraction

Panel A. Extraction scale				
Sample	Cases	ToHs	Edges	Edges/ToH
Long-run UK competition-harm corpus	1,301	2,806	12,081	4.31
Modern CMA subset	744	1,719	7,186	4.18
Non-CMA or historical remainder	557	1,087	4,895	4.50
Panel B. Canonical resolution of directed edges				
Sample	Edges	Both endpoints	One endpoint	Neither endpoint
Long-run UK competition-harm corpus	12,081	32.12%	49.95%	17.92%
Modern CMA subset	7,186	34.79%	51.25%	13.96%
Non-CMA or historical remainder	4,895	28.21%	48.05%	23.74%
Panel C. Sign-aware classification of resolved core-effect edges				
Sample	Core edges	Affirmative harm	Mitigating/no concern	Other
Long-run UK competition-harm corpus	2,215	45.96%	15.80%	38.24%
Modern CMA subset	1,347	46.70%	22.87%	30.44%
Non-CMA or historical remainder	868	44.82%	4.84%	50.35%

Notes: Cases in this table are cases with at least one extracted theory of harm. The modern CMA subset and the non-CMA/historical remainder partition the long-run extracted corpus. “Both endpoints” means that both source and target concepts in an extracted directed edge are resolved in the current ontology mapping. Core-effect edges are the subset of resolved edges that express a substantive mechanism relationship rather than a diagnostic, definitional, or metadata relationship. Affirmative harm means either an increase in a harmful object or a decrease in a beneficial constraint or welfare object. Percentages are provisional and will be updated after the ontology mapping, sample cut, and historical-comparability checks are settled.

Before making this a final paper result, settle the ontology mapping and keep the two-layer descriptive sample structure explicit. Reconcile the 1,339 substantive-document count in Table 1 with the 1,301 cases that currently have at least one extracted theory of harm. The long-run UK competition-harm corpus should be the main sample for historical questions. The modern CMA subset should be reported separately where the object depends on modern theory-of-harm language,

sign-aware paths, SLC outcomes, clearance/intervention status, or remedies. Keep the compact table provisional until the sample cut is settled: Panel A, extraction scale; Panel B, canonical resolution; Panel C, affirmative versus mitigating/non-concern core-effect edges. Keep the ranked mechanism relationships as a figure rather than adding a fourth dense table panel.

7.1 Mechanism composition over time

⟨placeholder⟩ Pending empirical output — mechanism-family composition across institutional corpora (MC/MMC, CC, OFT, CMA). Keep this descriptive. Report counts and shares only after deciding whether to condition on inquiry type, case type, and substantive-content filters. Avoid language implying a clean historical transition unless the comparison survives those checks.

This subsection will provide the first substantive description of how the mix of mechanisms changes across the long-run corpus. The relevant comparison is institutional-period by institutional-period, not a simple before/after claim. For the current draft, the important point is that mechanism-family shares will be interpreted only after checking whether differences reflect substantive reasoning, document form, or extraction yield.

7.2 Complexity and structure over time

⟨placeholder⟩ Pending empirical output — basic structure of the extracted object: ToHs per case, edges per ToH, and later motif/path measures after concept-level extraction. Use these measures to ask whether policy reasoning becomes more structured, more modular, or more specialized over time. Check whether edge counts reflect document length, report style, and prompt behavior before giving them substantive interpretation.

This subsection will ask whether the internal structure of policy reasoning changes over time. The simplest quantities are ToHs per case and edges per ToH. Richer quantities, such as repeated motifs or recurring templates, become useful only after the concept mapping is stable. The paper should report these measures as properties of written reasoning, not as direct measures of the complexity of the underlying market problem.

7.3 Within-family heterogeneity

⟨placeholder⟩ Pending empirical output — decomposition within horizontal unilateral effects by sub-mechanism (price, non-price, diversion, closeness, innovation). Frame and channel decomposition within vertical foreclosure (input vs customer foreclosure; upstream vs downstream harm). What broad family labels hide once we zoom in. Requires canonical concept assignments so sub-mechanism identity is stable across documents.

This subsection will show why broad mechanism-family labels are not enough. Two cases can both be coded as horizontal unilateral effects while relying on different channels: diversion, closeness,

capacity, innovation, quality, or bargaining. The graph representation matters here because it lets the paper describe the mechanism inside a family rather than only the family name.

7.4 Survival, entry, and exit of theories of harm

<placeholder> Pending empirical output — emergence of new motifs (canonical subgraphs) over time. Persistence and disappearance of specific theories of harm. Whether turnover differs by top mechanism family (e.g. do innovation-effects theories turn over faster than horizontal-unilateral ones?). Requires canonical motif counts from the concept-level policy extraction.

This subsection will connect the descriptive section to the origins section. If a mechanism appears, disappears, or becomes more common, the origins analysis can then ask whether that timing lines up with source-corpus appearances. The object to track is therefore not only the broad family, but the recurring mechanism component or template.

8 Origins of Competition-Policy Ideas

The origins section asks whether the mechanism components used in policy documents also appear in academic, legal, practitioner, or policy-source corpora, and when those components appear there. The section is descriptive in its first version: it measures overlap and timing, not causal influence. That distinction lets the paper say something concrete about where policy reasoning is located in the space of external ideas without claiming that any specific source caused any specific decision.

8.1 How much policy reasoning appears in academic mechanism graphs?

<placeholder> Pending empirical output — populate with canonical-triple results. Research question: how much of a policy theory-of-harm graph is already present in academic mechanism space? For each policy theory of harm or canonical ToH template, report the share of canonical triples matched in the *FrontierGraph* academic reservoir, the papers or contribution texts that contain each matched component, and the first-appearance year of those components. Use *FrontierGraph* as the baseline source reservoir. Use named contributions only as an interpretable benchmark or case study. Requires settled concept normalization and source-reservoir matching.

The baseline comparison will be policy-to-academic containment: for each policy theory of harm, what share of its mechanism components can be found somewhere in the academic mechanism reservoir? This is the broadest source-side test because it does not require naming a single canonical contribution in advance. Named contributions are useful as interpretable examples, but the main academic comparison should use the broader reservoir.

8.2 Diffusion timing from canonical triples

<placeholder> Pending empirical output — report timing patterns from canonical-triple origination. Research question: do policy mechanisms draw on recent academic components or on older absorbed components? For each frequently used policy mechanism component, report when it first appears in the academic reservoir and whether policy uses cluster soon after academic first appearance. Requires source-reservoir matching.

The timing exercise will separate old absorbed ideas from recent ideas. Some mechanisms may have academic antecedents decades before they become standard in policy language; others may appear in policy soon after they become prominent in source corpora. The paper should report this timing as precedence and absorption, reserving causal language for designs with a credible source of exposure variation.

8.3 Econ-heavy versus legal-heavy policy narratives

<placeholder> Priority empirical task — build the legal and practitioner source side in the same graph representation. The current source side is too economics-heavy to answer whether policy narratives draw more from economics, law, guidance, or practitioner synthesis. Once legal/reference material is extracted, compare economics, legal, practitioner, and policy-source matches using canonical triples or ToH-template graphs.

8.4 Idea traces over time within named contributions

<placeholder> Possible secondary analysis — time-series of matches to specific named contributions across institutional periods (MC / MMC / CC-OFT / CMA). For each contribution, when do matched mechanism components rise or fall? Do matches to the Chicago contributions (Bork, Posner, Stigler 1964, Demsetz) increase, decrease, or stay flat across the 2002 Enterprise Act discontinuity? Does the emergence of explicit theory-of-harm language coincide with a shift toward Motta and Baker, or does the shift predate the language? This subsection complements Section 8.2 by reporting per-contribution time series rather than only the first appearance of individual triples; full execution awaits per-case publication-year metadata for the modern CMA corpus.

<placeholder> Source expansion to consider — the named-contribution benchmark is interpretable but not complete. The next source-side expansion should consider legal scholarship, practitioner synthesis, agency guidance, competition-law textbooks, and older market-process or political-economy sources only where they supply concrete mechanism claims comparable to policy graphs. This matters especially for policy concepts that remain weakly covered by economics papers, such as legal standards, closeness/diversion, unilateral effects, switching costs, and institutionalized market-definition practice.

9 Policy Consequences and Economic Relevance

⟨placeholder⟩ Develop the causal/outcome analysis. First rebuild outcome coding and mechanism incidence from the current concept-level extraction; then define credible designs for how theories of harm, source matches, or ToH templates relate to intervention, remedies, and other policy outcomes.

The next step is to ask whether the measured reasoning matters for policy outcomes and economic stakes. The descriptive and origins sections can show what authorities say and where those mechanism components appear elsewhere. Here the paper will ask whether different mechanisms, templates, or source matches are associated with intervention, remedies, appeal outcomes, or economically larger cases. Until outcome coding and case metadata are rebuilt, the section should be read as the agenda for the next empirical layer rather than as a populated results section.

9.1 Which theories of harm matter for outcomes?

⟨placeholder⟩ Build the outcome dataset from the current concept-level extraction and cleaned case metadata. Minimum outputs: case-level outcome distribution, intervention/remedy rates by ToH template, and timing by institutional period. The causal version should define designs for whether mechanism templates, source matches, or exposure to source ideas predict intervention, remedies, appeal outcomes, or other policy consequences.

The first outcome analysis should stay close to observables: intervention, clearance, remedies, and appeal or revision where available. The unit should be the case, with ToH templates or mechanism incidence aggregated to the case level. That structure lets the paper ask whether some kinds of reasoning are more likely to appear in intervention cases before moving to stronger causal designs.

9.2 Back-of-the-envelope economic importance

⟨placeholder⟩ Develop the economic-importance calculation. Candidate denominators include affected-market turnover, transaction value, remedy scope, or sectoral exposure. The useful version links ToH templates or source traditions to economically meaningful policy stakes, with transparent coverage and missingness.

The purpose of this subsection is to keep the paper connected to economic magnitude. A mechanism that appears often in low-stakes cases and a mechanism that appears rarely in very large markets should not be interpreted in the same way. The useful version will report transparent coverage and missingness, then relate mechanism templates or source matches to transaction values, affected-market turnover, remedy scope, or sector exposure where those denominators are available.

10 Limits and Identification

⟨placeholder⟩ Turn this into the design agenda for causal claims. Distinguish overlap, precedence, exposure, and influence; then specify which event studies, guideline shocks, jurisdictional compar-

isions, or personnel/institutional changes could identify each step.

This section is a design placeholder until the descriptive, origins, and outcome analyses are populated. Section 7 documents the composition, complexity, and internal heterogeneity of competition-policy reasoning; Section 8.1 will characterise overlap, precedence, and source matches between policy reasoning and external reservoirs of mechanism claims; Section 9 will study how theories of harm, source matches, and ToH templates relate to policy outcomes. The causal version of the paper needs designs that separate shared mechanism content from exposure and influence.

That limitation is substantive rather than rhetorical. Academic work on merger analysis often responds to the same practical problems that authorities are already facing. Publication dates, personnel changes, legal reforms, and political shocks can also move together. Even a close match between a policy mechanism and an academic passage establishes shared mechanism content, not individual exposure or causal influence.

The measured mechanism graph supplies stable mechanism units, dates, and source matches that could support event studies around distinctive academic contributions, discontinuity designs around legal reforms, or cross-jurisdictional comparisons around foreign guideline changes. Those designs would require additional assumptions and, in several cases, additional corpora. They are therefore treated as later analyses rather than as results of this paper.

The task here is to spell out what each design would identify: descriptive overlap, chronological precedence, likely exposure, or causal influence.

11 Conclusion

<placeholder> Draft after the main empirical sections are populated. The conclusion should answer the substantive questions: where competition-policy reasoning appears to come from, which mechanisms persist or change, and how those changes relate to policy outcomes. Measurement should appear as the enabling contribution, not the closing claim.

The conclusion will return to the paper’s substantive question: how competition-policy reasoning changes, where its mechanism components appear outside policy, and whether those components matter for policy outcomes. The graph representation is the enabling measurement contribution. The paper should close on what that measurement teaches us about competition-policy ideas, not on the machinery used to extract them.

A Data sources and acquisition details

This appendix documents sourcing details and per-corpus provenance for every dataset listed in §5.1. Table 3 summarises the acquisition route, source format, and status for each corpus.

Table 3: Per-corpus acquisition status

Corpus	Acquisition route	Format	Status
<i>A. UK competition policy corpus</i>			
Historical inquiries (MC + MMC)	CC archive and gov.uk archive	Multi-PDF chapter sets	Acquired; concept extraction complete
CC “new” inquiries	Same scrape; filtered to post-1999 inquiries	Single multi-PDF case folders	Acquired; concept extraction complete
OFT phase-1 decisions	Web-scrape of OFT archive on gov.uk archive	Single decision pages	Acquired; concept extraction complete
Modern CMA cases	Pre-existing scrape from CMA register on gov.uk	Pre-extracted plain text	Acquired; concept extraction complete
<i>B. Named contributions corpus</i>			
Primary-text contributions	Direct download from publisher / archive	PDF or plain text	Acquired; benchmark extraction present
Substitute contributions	Curated publicly-available articles by same author	PDF	Acquired; benchmark extraction present
Paraphrase-based contributions	Curator-prepared paraphrase from secondary literature	Plain text	Generated; benchmark extraction present
<i>C. Additional comparison resources</i>			
1980s MMC reports gap	TNA reading-room visit; possible British Library scan; HMSO partial	Print only; OCR required	Not started
EU Commission (DG COMP)	Commission archive pull	PDF with case index	Acquired, not extracted
US DoJ antitrust	Web-scrape of DoJ public archive	Mixed PDF and HTML	Acquired, not extracted
FrontierGraph academic reservoir	Parallel extracted economics corpus	Structured paper-edge corpus	External, read-only
OpenAlex T11043 top-up	OpenAlex topic query	Metadata and abstracts	Parked
Legal reference corpus	Manual curation: law-school syllabi + treatises	PDF and text sources	Not started

Notes: Status values distinguish document acquisition from paper-facing concept extraction. *Acquired; concept extraction complete* = raw documents or text are present and the concept-level ToH graphs have been extracted, with final reporting still subject to sample-cut and comparability checks. *Benchmark extraction present* = structured source-side extraction exists for interpretive benchmark use. *Acquired, not extracted* = raw documents present, no structured outputs yet. *Not started* = no raw documents acquired. The 1980s MMC reports gap is a documented sourcing problem rather than an extraction problem: the MMC reports of 1980–1989 (corresponding to inquiries numbered 122–263 in the digitised series) exist as physical documents at the National Archives but were not part of the Competition Commission’s pre-2014 digitisation effort.

UK policy corpus extraction summary. The four UK sub-corpora have usable document text and earlier structured extractions, but the paper-facing extraction is the concept-level ToH graph pass described in §6.1. Historical and CC “new” cases use concatenated chapter-PDF text; OFT cases use single-page HTML stripped of markup; modern CMA cases use pre-extracted plain text. Corpus-level ToH, edge, mechanism-composition, and outcome tables should be rebuilt from the concept-level pass.

Named contributions corpus extraction summary. The 21 contributions were extracted in two phases. Phase 1 used the extraction procedure on primary-text or substitute texts. Phase 2 used curator-prepared paraphrases from secondary literature and then applied the same extraction schema. Each paraphrase-based contribution carries a provenance note recording the secondary-literature basis.

Planned-corpus acquisition status. DG COMP. The EU Commission merger-decision archive contains 9,695 case directories with PDF documents and a case index. Extraction has not started; the UK extraction procedure is compatible with minor adjustments for French and German text in older cases.

US DoJ. The DoJ public archive was scraped to identify approximately 2,200 antitrust cases. Document-level acquisition is partial; some older case documents require alternative acquisition routes.

Frontier economics. The main academic reservoir is the external *FrontierGraph* corpus, with approximately 228,000 economics papers and 1.24 million canonical-resolved edges. The OpenAlex topic T11043 (“Merger and Competition Analysis”) was also queried in May 2026, but it is parked as an optional top-up rather than treated as the primary reservoir.

Legal reference corpus. Not yet acquired. Planned strategy: curated law-school antitrust syllabi (Yale TAP, Harvard, Chicago), competition-law treatises (Whish and Bailey on EU competition law; Hovenkamp on US antitrust; Goyder on EC competition law), and selected book chapters with substantial mechanism content. Acquisition will run alongside the next round of frontier-corpus work.

1980s MMC reports gap. The Competition Commission’s pre-2014 digitisation effort included MMC reports through 1979 (inquiry numbers 1–121) and from 1990 onwards (numbers 264 onwards). The intervening 1980–1989 reports (approximately 142 inquiries by the numbered slug range) exist as bound print documents at the National Archives and the British Library. Filling the gap requires either an in-person scanning visit or a commissioned scanning service. The gap is treated as a known limitation in §5.2.

B Named-contribution source corpus

Table 4 reports the source corpus used for the named-contribution comparison. “Basis” indicates whether the extraction uses the primary text, a substitute text by the same author on the same intellectual territory, or a curator-prepared paraphrase based on secondary literature.

Table 4: Named contributions in antitrust economics

Year	Contribution	Core idea	Basis
1937	Coase	Nature of the firm; transaction-cost foundation	paraphrase
1940	Arnold	New Deal antitrust enforcement	substitute
1942	Schumpeter	Creative destruction	primary
1948	Bain	Conditions of entry; structure-conduct-performance	substitute
1962	Arrow	Appropriability of invention	primary
1964	Stigler	Theory of oligopoly	primary
1966	Bork	Legislative intent; consumer-welfare standard	substitute
1968	Neal Report	Structural antitrust policy	primary
1971	Stigler	Theory of economic regulation	primary
1973	Demsetz	Efficient structure hypothesis	primary
1979	Posner	Chicago school antitrust analysis	primary
1979	Williamson	Transaction-cost economics; governance modes	substitute
1983	Salop–Scheffman	Raising rivals’ costs	paraphrase
1985	Hovenkamp	Post-Chicago antitrust	substitute
1988	Tirole	Game-theoretic synthesis of industrial organization	paraphrase
1990	Whinston	Tying, foreclosure, exclusion	paraphrase
2004	Motta	Competition policy textbook synthesis	primary
2010	Farrell–Shapiro	Upward pricing pressure / GUPPI	paraphrase
2017	Khan	Amazon’s antitrust paradox; neo-Brandeisian critique	paraphrase
2018	Wu	Curse of bigness; political-economy critique	paraphrase
2019	Baker	Modern enforcement synthesis	primary

C Historical extraction validation

For documents that predate the theory-of-harm terminology (primarily Monopolies Commission and MMC reports from the 1950s to early 1990s), we run two validity checks on the extraction output to assess comparability with modern CMA decisions. First, a stratified audit sample of historical reports is hand-coded by a trained reader for the presence, identity, and structure of mechanism claims, and the audit coding is compared against the automated extraction using precision, recall, and edit-distance measures on the set of returned ToHs and edges. Second, a post-hoc linguistic-style audit checks whether historical extractions differ systematically from modern extractions on observable surface features—mean edge count per ToH, node-label length, relation-category mix, and share of edges flagged as *ambiguous* link form. Results of both checks will be reported in a later draft once the validation audit is complete.

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